

Table 2. Summary statistics of the variables of the dataset (6,446 articles)

Variable	Mean	Std dev.	Min	Max
<i>Outcome</i>				
Impact	1.545	2.861	0	91.917
<i>Quality</i>				
Peers' score				
Limited (0.2)	2.2%			
Acceptable (0.6)	16.6%			
Good (0.8)	49.0%			
Excellent (1)	32.1%			
Short-term impact	1.280	1.739	0	29.564
<i>Byline</i>				
N. authors	5.508	6.718	1	99
Avg authors impact	0.967	0.829	0	25.845
Foreign	0.369	0.483	0	1
<i>Content and venue</i>				
Open Access	0.318	0.466	0	1
N. pages	10.955	8.355	1	95
Journal IF	33.119	18.25	2	196
<i>Bibliography</i>				
N. references	2.215	2.02	0	15.792
% references in WoS	68.711	22.328	2.941	100
% self-cites	22.188	19.118	0	100
Cited articles age	7.021	2.644	0	21
Cited articles avg impact	10.285	22.26	0.035	520.765

4.2.2 The statistical model

We exploit a linear random effects model to evaluate the extent of the association of the above-mentioned non-scientific features via-à-vis the quality of a given publication with its scholarly impact (e.g., Rabe-Hesketh & Skrondal, 2022). Specifically, for an article i published in journal j , the model is:

$$Y_{ij} = \alpha + \delta Z_{ij} + \beta_1 X_{1ij} + \dots + \beta_p X_{pij} + u_j + e_{ij} \quad [1]$$

where Y_{ij} is the outcome variable, i.e., the article's (normalised) impact, Z_{ij} is a measure of the article's quality and $X_{1ij}, \dots, X_{pij}$ are the article's non-scientific factors.

Model [1] includes a random effect u_j representing the unobserved factors for journal j and a residual error e_{ij} . Both u_j and e_{ij} are assumed to be normal random variables with zero mean and unknown standard deviations denoted with σ_u and σ_e , respectively. The model is fitted using the *xtreg* command of Stata 18 with the *re* option (StataCorp., 2023).

A preliminary analysis showed that the relationships are linear on the logarithmic scale for all "impact" variables with right-skewed distributions. For this reason, those variables are log-transformed. Values equal to zero are replaced with the minimum positive value before computing the logarithm.

5. Results

We fitted three models, each including all the relevant non-scientific factors in Table 2. The response variable is the logarithm of the article's impact. The explanatory variables

based on a measure of impact are also logarithmic. The first model includes only non-scientific factors as predictors, the second model adds the “quality” of the article in terms of the score assigned by the peers, and the third model substitutes the quality scores with the short-term impact as measured by the early citations. The results are reported in Table 3. The R-squared, a measure of predictive ability, is 0.248 when considering only non-scientific factors. It increases slightly to 0.260 when including the score assigned by peers and substantially to 0.485 when adding early citations. Therefore, the value added by the peers is marginal and much smaller than the contribution given by an easily and cheaply calculated bibliometric indicator of short-term impact.

All explanatory variables are statistically significant at the 0.01 level, except for open access in all models and the number of authors in the third one. Moreover, in the second model, the categories 0.2 and 0.6 of the score assigned by peers do not achieve statistical significance (p -values equal to 0.8909 and 0.0650, respectively): This means that, when controlling for non-scientific factors, articles with scores of 0.2 and 0.6 do not exhibit statistically different impacts compared to those with a score of 0.8 (baseline). However, the top score, namely 1, has a highly significant effect (p -value < 0.0001), meaning that reviewers better identify excellent rather than poor papers. Specifically, an article scoring 1 by peers has, other things being equal, a long-term scholarly impact about 28% greater than an article scoring 0.8. On the other hand, the coefficient of the short-term impact is an elasticity measure; thus, a 1% increase in the early citations is associated with a 0.446% increase in the long-term impact.

Non-scientific factors consistently influence the outcome of all models. Their effects tend to be smaller in magnitude in the model with the early citations. The reason is that the impact measured by early citations is a mediator for the long-term impact, absorbing part of the effects; nonetheless, it is remarkable that most non-scientific factors are relevant, even controlling for the early citations. Specifically, as for the byline, the number of authors is not statistically significant, whereas an increase of 1% in the average author’s impact is associated with a 0.246% increase in the long-term impact, and the presence of a foreign author is associated with a 7.2% increase. As for content and venue, open access is not statistically significant, whereas one more page length is associated with 1.1% more impact, and an increase of 1% in the journal IF corresponds to 0.267% more impact. As for features related to the publication’s bibliography, we found different patterns: the effect on the long-term impact is positive for the number of references (+0.4% for one additional reference) and the average impact of cited articles (+0.101% for 1% more average impact). On the other hand, the relationship is negative for the cited articles’ age (-2.3% for one additional year), the percentage of self-cites (-0.4% for one additional percentage point), and the percentage of cited articles in WoS (-0.4% for one additional point). The estimated coefficients generally align with expectations, except for the negative coefficient associated with the percentage of cited articles in WoS. This unexpected result may warrant further investigation. However, it should be noted that this effect is adjusted for the other explanatory variables, particularly the bibliometric ones.

Considering the residual variances, it is worth noting that inserting the peers’ score in the second model has negligible consequences. On the other hand, inserting the early impact in the third model leaves the journal-level variance unchanged while substantially reducing the article-level variance; thus, the fraction of residual variance due to journals rises from 19.6% to 27.3%. It is remarkable that in the third model, 27.3% of the residual variance is at the journal level, i.e., more than one-fourth of the variance due to unmeasured factors is attributable to the journal. This implies a residual correlation of

0.273 among articles published in the same journal, even controlling for the journal IF. Therefore, the journal plays an important role in determining the long-term impact beyond the value of its IF.

The analysis has been replicated at the disciplinary area level of the articles. Table 4 shows the R-squared for the three fitted models. The overall pattern is confirmed in all areas: once the non-scientific factors are considered, the score assigned by peers barely improves the prediction of the long-term impact, whereas the short-term impact provides a substantial improvement, especially in Physics and Biology.

Table 3. Point estimates (standard errors in parenthesis) for three linear random effects models for the logarithm of the article's impact controlling for non-scientific factors (6,446 articles in 1,211 journals)

	Quality/Impact		
	None	Peer review	Short-term impact
R-squared	0.248	0.260	0.485
Intercept	-0.848 (0.095)	0.095 (-0.792)	-0.792 (0.096)
<i>Byline</i>			
N. authors	0.007 (0.002)	0.007 (0.002)	0.003 (0.002)
Avg authors' impact (log)	0.381 (0.018)	0.364 (0.018)	0.246 (0.015)
Foreign	0.127 (0.027)	0.112 (0.027)	0.072 (0.023)
<i>Content and venue</i>			
Open Access	0.076 (0.035)	0.058 (0.035)	0.019 (0.030)
N. pages	0.014 (0.002)	0.011 (0.002)	0.011 (0.002)
Journal IF (log)	0.668 (0.060)	0.593 (0.061)	0.267 (0.056)
<i>Bibliography</i>			
N. references	0.008 (0.001)	0.008 (0.001)	0.004 (0.001)
% references in WoS	-0.004 (0.001)	-0.004 (0.001)	-0.004 (0.001)
% self-cites	-0.003 (0.001)	-0.004 (0.001)	-0.004 (0.001)
Cited articles age	-0.062 (0.005)	-0.061 (0.005)	-0.023 (0.004)
Cited articles avg impact (log)	0.165 (0.013)	0.163 (0.013)	0.101 (0.011)
<i>Quality</i>			
Peers' score			
Limited (0.2)		-0.012 (0.085)	
Acceptable (0.6)		-0.066 (0.036)	
Good (0.8) – baseline		---	
Excellent (1)		0.280 (0.030)	
Early citations (log)			0.446 (0.008)
<i>Residual variances</i>			
Level 2 (journal)	0.212	0.211	0.217
Level 1 (article)	0.869	0.854	0.579
Fraction of level 2 variance	0.196	0.198	0.273

Table 4. Sample size and values of the R-squared for three linear random effects models for the logarithm of the article's impact controlling for non-scientific factors, by disciplinary area

Disciplinary area	Obs	Quality/Impact		
		None	Peer review	Short-term impact
1 - Mathematics and computer science	683	0.29	0.30	0.41
2 - Physics	1184	0.33	0.37	0.66
3 - Chemistry	912	0.24	0.25	0.52
4 - Earth sciences	536	0.23	0.24	0.54
5 - Biology	1331	0.27	0.30	0.62
7 - Agricultural and veterinary sciences	627	0.25	0.25	0.42
8 - Civil engineering	307	0.21	0.23	0.42
9 - Industrial and information engineering	921	0.21	0.22	0.38
Overall	6446	0.25	0.26	0.49

6. Discussion and conclusions

The prevailing view among the majority of research assessment scholars and practitioners is that peer-review evaluation stands as the gold standard, while citation-based evaluation serves as a quicker and more economical surrogate. Consequently, considerable scholarly attention has been directed towards assessing how effectively bibliometric evaluation approximates peer review, with potential for substitution or complementary use across various domains: i) individual scientific publications (Bertocchi, Gambardella, Jappelli, Nappi, & Peracchi, 2015; Bornmann & Leydesdorff, 2013); ii) individual researchers (Vieira & Gomes, 2018; Cabezas-Clavijo, Robinson-García, Escabias, & Jiménez-Contreras, 2013), and iii) research institutions (Pride & Knoth, 2018; Franceschet & Costantini, 2011).

However, the underlying rationale for the presumption of peer review's superiority over evaluative bibliometrics remains elusive.

On the empirical front, Abramo, D'Angelo, and Reale (2019) presented evidence that early citations offer better predictions of the long-term scholarly impact of scientific publications compared to peer-review quality scores. The present study reinforces these findings, even after factoring in the influence of non-scientific factors on impact.

Similarly, theoretical foundations are also wanting. In recent years, policymakers have rightly shifted their focus towards assessing the social impact of research rather than merely its quality. While research output quality serves as a means to an end, the ultimate goal of research activity invariably remains its societal impact. This impact is influenced partly by research institutions and partly by the capacity of both the industrial and public sectors to incorporate research findings into enhanced products and processes. Concerning institutional responsibilities, a clear division between production and dissemination - between researchers and technology licensing offices - is essential. Researchers must not only generate quality results but also ensure their swift dissemination within the scholarly community, utilising non-scientific factors responsibly to enhance scholarly impact.

Recent national evaluation initiatives have partly embraced these new priorities, mandating institutions to demonstrate the societal impact of their research by submitting relevant case studies. However, by retaining peer-review evaluation of publications as the primary metric, these assessments continue to prioritise research quality over scholarly impact. Peers, often experts in specific fields, face challenges in assessing the broader scientific implications of research beyond their expertise. Recent studies reveal that citations to publications predominantly originate from diverse domains (Abramo & D'Angelo, 2024).

While the quality of scientific products typically plays a significant role in scholars' choices, other factors also exert influence. As demonstrated in this study concerning scientific articles, factors such as the reputation of authors and the publication venue, alongside quality, shape their utilisation by the scientific community for advancing knowledge. Additionally, factors like the average impact of cited works and multinational authorships contribute to a lesser extent. Embracing bibliometric methods, which effectively measure scholars' citations as "consumer purchases," could advance policymakers' shared objectives. Replacing "evaluation of research quality" with "evaluation of research impact" in technical parlance would foster greater clarity among professionals and stakeholders regarding evaluation objectives.